Financial Impacts & Workflows

Complete reference for all financial transactions, auto-posting rules, and accounting workflows in the HR system.

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1. Sales Revenue

Trigger: Sales log submitted by sales agent (`POST /api/sales/log`)

Account Mapping:

Direction	Account	Amount
Debit	Accounts Receivable	grand_total
Credit	Sales Revenue	grand_total - vat_amount
Credit	VAT Payable	vat_amount (if > 0)

Workflow:

```
Sales Agent submits log
       ↓
Status: "submitted"
       ↓
Auto-post fires (non-blocking)
       ↓
Journal Entry created (source: "sales_log")
       ↓
Notifications sent to managers & org admins
```

Auto-Posting Rule:

- Flag: `accounting_settings.auto_post_sales`
- Posts immediately on submission
- Template: `src/lib/accounting/templates/sales.ts`

2. Payroll Accrual

Trigger: Manual trigger via `POST /api/accounting/payroll/auto-post` with target month

Account Mapping:

Direction	Account	Amount
Debit	Salary Expense	total_net + advance_deductions
Credit	Salary Payable	total_net + advance_deductions
Debit	Salary Payable	advance_deductions (if > 0)
Credit	Employee Advances	advance_deductions (if > 0)

Components Captured:

- `base_salary`
- `total_allowances` (included in total_net)
- `total_deductions` (parsed from deductions_details JSON)
- `advance_deductions` (items containing "advance" or "سلفة")
- `employer_gosi` & `employee_gosi` (tracked in payslips table)

Workflow:

```
Payslips created (status: "draft")
       ↓
HR reviews & approves payslips
       ↓
Status: "approved"
       ↓
Admin triggers monthly auto-post
       ↓
All approved payslips aggregated
       ↓
Single Journal Entry created (entry_date: 28th of month)
       ↓
Source: "payroll"
```

Deduction Parsing Logic:

- Stored in `payslips.deductions_details` as JSON array
- Advance detection: checks if label contains "advance" (case-insensitive) or "سلفة" (Arabic)
- Aggregated and offset against salary payable in GL entry

Auto-Posting Rule:

- Flag: `accounting_settings.auto_post_payroll`
- Requires all payslips to be in "approved" status
- Template: `src/lib/accounting/templates/payroll.ts`

3. Employee Advances

Trigger: Advance request approved or partially approved (`PATCH /api/advance-requests`)

Account Mapping:

Direction	Account	Amount
Debit	Employee Advances Receivable	approved_amount
Credit	Bank/Cash Account	approved_amount

Workflow:

```
Employee submits advance request (with signature)
       ↓
Status: "pending"
       ↓
HR admin/manager reviews
       ↓
Decision: approved / partial_approved / rejected
       ↓
If approved:
  ├── Financial balance record created (status: "unpaid")
  ├── Auto-post fires (non-blocking)
  ├── Journal Entry created (source: "advance")
  └── Notification sent to employee
       ↓
Deductions applied via subsequent payroll cycles
       ↓
Financial balance → "paid" when fully recovered
```

Auto-Posting Rule:

- Flag: `accounting_settings.auto_post_advances`
- Posts immediately upon approval (non-blocking)
- Stores `journal_entry_id` reference in `advance_requests` table
- Template: `src/lib/accounting/templates/advance.ts`

4. Asset Capitalization & Depreciation

4A. Asset Capitalization

Trigger: `POST /api/accounting/assets/capitalize` (employee asset assigned with purchase price)

Account Mapping:

Direction	Account	Amount	Condition
Debit	Fixed Assets	purchase_price	Always
Credit	Bank/Cash	purchase_price	If paid_immediately = true
Credit	Accounts Payable	purchase_price	If paid_immediately = false

Workflow:

```
Asset assigned to employee with purchase price
       ↓
Capitalization triggered
       ↓
Journal Entry created (source: "asset_capitalization")
       ↓
capitalized_journal_id stored in employee_assets table
```

4B. Monthly Depreciation

Trigger: Batch job `POST /api/accounting/assets/run-depreciation` with target month

Depreciation Calculation (Straight-Line):

```
Monthly Depreciation = (purchase_price - salvage_value) / useful_life_months
```

Account Mapping:

Direction	Account	Amount
Debit	Depreciation Expense	monthly_depreciation_amount
Credit	Accumulated Depreciation	monthly_depreciation_amount

Workflow:

```
Admin triggers monthly depreciation batch
       ↓
All active assets with purchase_price > 0 selected
       ↓
Per-asset depreciation calculated
       ↓
Single combined Journal Entry for all assets
       ↓
last_depreciation_date updated on each asset
       ↓
Source: "asset_depreciation"
```

Auto-Posting Rule:

- Flag: `accounting_settings.auto_post_assets`
- One combined JE for all depreciation in the period
- Schema fields: `purchase_price`, `useful_life_months`, `salvage_value`, `depreciation_method`

5. End of Service Benefits (EOSB)

5A. Monthly EOSB Accrual

Trigger: Batch job `POST /api/accounting/eosb/run-accrual` with target month

EOSB Calculation Logic (Saudi Labor Law):

```
Daily Rate = Base Salary / 30

Full Service Entitlement:
  Years 1-5:  15 days per year
  Year 5+:   15 days (first 5 years) + 30 days per additional year
  Formula:    (5 × 15) + ((years - 5) × 30) days × daily_rate

Resignation (Reduced):
  < 2 years:   0 (no benefit)
  2-5 years:   1/3 of calculated amount
  5-10 years:  2/3 of calculated amount
  > 10 years:  Full amount

Termination / End of Contract:
  Full calculated amount
```

Account Mapping:

Direction	Account	Amount
Debit	EOSB Expense	monthly_accrual
Credit	EOSB Provision	monthly_accrual

monthly_accrual = current_month_EOSB - previous_month_EOSB

Workflow:

```
Admin triggers monthly EOSB accrual batch
       ↓
All active employees with hiring_date selected
       ↓
Per-employee EOSB calculated based on tenure
       ↓
Monthly increment = current_total - last_cumulative
       ↓
Recorded in eosb_accruals table
       ↓
Combined Journal Entry (entry_date: 28th of month)
       ↓
Source: "eosb"
```

Tracking (eosb_accruals table):

- `years_of_service` , `daily_rate`
- `monthly_accrual` , `cumulative_accrual`
- `journal_entry_id` reference
- UNIQUE constraint prevents duplicate accruals per employee/month

5B. EOSB Settlement

Trigger: Employee terminated via `POST /api/accounting/eosb/settle`

Account Mapping:

Direction	Account	Amount	Condition
Debit	EOSB Provision	min(accruedAmount, settlementAmount)	If accrued > 0
Debit	EOSB Expense	settlementAmount - accruedAmount	If settlement > accrued
Credit	Bank/Cash	settlementAmount	Always

Workflow:

```
Employee termination processed
       ↓
Settlement amount calculated based on termination_type
       ↓
Compared against cumulative accrual
       ↓
Difference (if any) charged to EOSB Expense
       ↓
Accrued portion reversed from EOSB Provision
       ↓
Payment issued from Bank/Cash
       ↓
Source: "eosb_settlement"
```

Auto-Posting Rule:

- Flag: `accounting_settings.auto_post_eosb`

- Applies to both monthly accrual and settlement

6. Leave Encashment

Trigger: `POST /api/accounting/leave-encashment` (typically at termination or leave payout)

Calculation:

```
Encashment Amount = vacation_remaining_days × (base_salary / 30)
```

Account Mapping:

Direction	Account	Amount
Debit	Leave Encashment Expense	encashment_amount
Credit	Salary Payable	encashment_amount

Note: Leave Encashment Expense defaults to Salary Expense account if `default_leave_encashment_id` not set.

Workflow:

```
Leave encashment requested (usually at termination)
       ↓
Employee's base_salary and vacation_remaining looked up
       ↓
Encashment calculated
       ↓
Journal Entry created with leave days in description
       ↓
Source: "leave_encashment"
```

7. GOSI Payment

Trigger: Manual entry via `POST /api/accounting/gosi/record-payment` with month and amount

Account Mapping:

Direction	Account	Amount
Debit	GOSI Payable	total_amount
Credit	Bank/Cash	total_amount

Workflow:

```
GOSI contributions accrued during payroll posting
  ├── employee_gosi (employee contribution)
  └── employer_gosi (employer contribution)
       ↓
Actual payment made to GOSI authority
       ↓
Admin records payment manually
       ↓
Journal Entry created (source: "gosi_payment")
```

8. Invoice Issuance

Trigger: Invoice status changes to "issued" via `POST/PATCH /api/accounting/invoices`

Account Mapping:

Direction	Account	Amount
Debit	Accounts Receivable	total_amount
Credit	Sales Revenue	total_amount - vat_amount
Credit	VAT Payable	vat_amount (if > 0)

Revenue recognized by line item - each line can have its own `revenue_account_id` . Lines using the same account are aggregated.

Workflow:

```
Invoice created (status: "draft")
       ↓
Line items added with amounts, VAT, account mappings
       ↓
Invoice issued (status: "issued")
       ↓
Journal Entry created automatically
       ↓
journal_entry_id stored on invoice record
       ↓
Source: "invoice"
```

9. Expense Recording & Payment

9A. Expense Accrual

Trigger: Expense approved via `POST/PATCH /api/accounting/expenses`

Account Mapping:

Direction	Account	Amount	Condition
Debit	Expense Category	amount_before_vat	Always
Debit	VAT Recoverable	vat_amount	If VAT > 0
Credit	Accounts Payable	total_amount	If not paid immediately
Credit	Bank/Cash	total_amount	If paid immediately

Workflow:

```
Expense created (status: "draft")
       ↓
Submitted for approval (status: "pending_approval")
       ↓
Finance manager approves (status: "approved")
       ↓
Journal Entry created
       ↓
Source: "expense"
```

9B. Expense Payment

Trigger: Payment recorded for approved expense via `/api/accounting/payments`

Account Mapping:

Direction	Account	Amount
Debit	Accounts Payable	payment_amount
Credit	Bank/Cash	payment_amount

Status Lifecycle:

```
draft → pending_approval → approved → paid
```

10. Payment Recording

Unified template covering all payment types:

Payment Type	Direction	Debit	Credit
customer_receipt	Inbound	Bank/Cash	Accounts Receivable
vendor_payment	Outbound	Accounts Payable	Bank/Cash
payroll_payment	Outbound	Salary Payable	Bank/Cash
employee_advance	Outbound	Employee Advances	Bank/Cash

Payment Type	Direction	Debit	Credit
manual	Flexible	Custom	Custom

Workflow:

```
Payment submitted with type & counterpart
       ↓
Account mappings resolved based on payment type
       ↓
Journal Entry created
       ↓
Source: "payment_{type}"
```

11. Financial Balance Tracking

Tracks employee-level financial obligations across time.

Balance Types:

- `deficit` - Overpayment / negative balance
- `loan` - Long-term employee loan
- `advance_payment` - Approved advance request

Status:

- `unpaid` - Balance outstanding
- `paid` - Fully settled

Lifecycle:

```
Advance approved → balance created (status: "unpaid")
       ↓
Deductions applied each payroll cycle
       ↓
Balance reduced progressively
       ↓
Fully recovered → status: "paid"
```

12. Auto-Posting Configuration

Flags in `accounting_settings` table:

Flag	Controls	Default
`auto_post_sales`	Sales revenue posting on log submission	true
`auto_post_payroll`	Payroll accrual posting	true
`auto_post_advances`	Advance disbursement posting on approval	true
`auto_post_assets`	Asset capitalization & depreciation	true
`auto_post_eosb`	EOSB accrual & settlement	true

Behavior When Disabled:

- Business operations proceed normally
- No GL entries created automatically
- Manual journal entries can still be posted
- Can be re-enabled and historical postings triggered manually

13. Journal Entry Structure

Journal Entry (`accounting_journal_entries`):

Field	Description
`entry_no`	Sequential (e.g., JE-000001)
`entry_date`	Transaction date
`status`	draft / pending_approval / posted / reversed / cancelled
`source_module`	Origin system (payroll, sales_log, etc.)
`source_id`	Reference to source record
`total_debit`	Must equal total_credit

Field	Description
`total_credit`	Must equal total_debit
`currency`	SAR (Saudi Riyal)
`created_by`	Audit: who created
`approved_by`	Audit: who approved
`posted_by`	Audit: who posted
`reversed_by`	Audit: who reversed

Journal Lines (`accounting_journal_lines`):

Field	Description
`account_id`	GL account reference
`debit`	Debit amount (never both debit & credit)
`credit`	Credit amount (never both debit & credit)
`description`	Line-level detail
`cost_center_id`	Optional allocation
`employee_id`	Context tracking
`department_id`	Context tracking
`location_id`	Context tracking

14. Approval & Authorization

Journal Entry Status Flow:

```
draft
  ↓ (submit for approval)
pending_approval
  ↓ (finance manager approves)
posted (immutable - cannot be edited)
  ↓ (if error discovered)
reversed (creates reversing entry, original stays intact)

draft → cancelled (if never submitted)
```

Role Permissions:

Action	Roles
View	admin, finance_manager, accountant, finance_viewer
Manage (create/edit)	admin, finance_manager, accountant
Post	admin, finance_manager, accountant
Settings	admin, finance_manager

15. Error Handling

Non-Blocking Auto-Post Pattern:

All auto-posting operations are non-blocking - business operations continue regardless of GL posting success.

```
Business action triggers auto-post
       ↓
Attempt GL entry creation
       ↓
Success → journal_entry_id stored on source record
       ↓
Failure → logged to accounting_source_postings (status: "failed")
         Business action still succeeds
         Can be retried manually later
```

Failure Tracking (`accounting_source_postings`):

- Records all auto-post attempts
- Status: "success" or "failed"
- Error details captured for debugging
- If logging itself fails, silently ignored (truly non-blocking)

16. Source Modules

Every journal entry tracks its origin for audit trail:

Source Module	Origin
`sales_log`	Sales revenue recognition
`payroll`	Monthly payroll accrual
`advance`	Advance disbursement
`asset_capitalization`	Fixed asset purchase
`asset_depreciation`	Monthly depreciation batch
`eosb`	EOSB monthly accrual
`eosb_settlement`	EOSB terminal settlement
`leave_encashment`	Unused leave payout
`gosi_payment`	Social insurance payment
`invoice`	Customer invoice issuance
`expense`	Expense accrual
`expense_paid`	Expense payment
`payment_customer_receipt`	Cash inbound
`payment_vendor_payment`	Vendor cash outbound
`payment_payroll_payment`	Salary payment
`payment_employee_advance`	Advance disbursement payment

Source Module	Origin
`payment_manual`	Manual journal entry
`manual`	User-created journal entry

17. Reporting

Available Financial Reports:

Report	Description
Trial Balance	All account balances at a point in time
General Ledger	Detailed transactions per account
Balance Sheet	Assets, Liabilities, Equity snapshot
P&L Statement	Revenue vs Expenses for a period
Cash Flow Statement	Cash inflows and outflows
AR Aging	Outstanding receivables by age
AP Aging	Outstanding payables by age
Employee Advances	Outstanding employee balances
Payroll Cost	Salary & benefits expense breakdown
VAT Report	VAT collected vs recoverable
Cost Center P&L	Profit/Loss by cost center
Budget Variance	Actual vs budgeted amounts

Cost Center Dimensions:

- Department
- Location
- Project
- Sales Company
- Branch

- Channel
- Custom

Financial Transactions Unified View (`financial_transactions` table):

- Aggregates all GL postings
- Classifies by `transaction_type` (invoice_issued, salary_accrual, etc.)
- Classifies by `direction` (inflow, outflow, obligation, adjustment)
- Enables cross-module reporting

18. Default Account Mapping

All default accounts configured in `accounting_settings`:

Setting Key	Purpose
`default_cash_account_id`	Cash on hand
`default_bank_account_id`	Primary bank account
`default_accounts_receivable_id`	Customer receivables
`default_accounts_payable_id`	Vendor payables
`default_sales_revenue_id`	Sales income
`default_vat_payable_id`	VAT output (collected)
`default_vat_receivable_id`	VAT input (recoverable)
`default_salary_expense_id`	Salary costs
`default_salary_payable_id`	Salaries owed to employees
`default_employee_advances_id`	Advances receivable from employees
`default_asset_account_id`	Fixed assets
`default_depreciation_expense_id`	Depreciation costs
`default_accumulated_depreciation_id`	Contra-asset for depreciation
`default_gosi_payable_id`	GOSI contributions owed

Setting Key	Purpose
`default_gosi_expense_id`	GOSI employer costs
`default_eosb_expense_id`	EOSB periodic expense
`default_eosb_provision_id`	EOSB liability provision
`default_leave_encashment_id`	Leave payout expense
`default_allowances_expense_id`	Employee allowances

19. File Reference

Component	Path
Auto-posting Engine	`src/lib/accounting/auto-posting.ts`
Posting Engine (Core)	`src/lib/accounting/posting-engine.ts`
Sales Template	`src/lib/accounting/templates/sales.ts`
Payroll Template	`src/lib/accounting/templates/payroll.ts`
Advance Template	`src/lib/accounting/templates/advance.ts`
Expense Template	`src/lib/accounting/templates/expense.ts`
Invoice Template	`src/lib/accounting/templates/invoice.ts`
Payment Template	`src/lib/accounting/templates/payment.ts`
Accounting Service	`src/lib/services/accounting.ts`
Main API Route	`src/app/api/accounting/[...segments]/route.ts`
Advance Requests API	`src/app/api/advance-requests/route.ts`
Sales Log API	`src/app/api/sales/log/route.ts`
Reports Library	`src/lib/accounting/reports/`
Financial Reporting Schema	`db/schema_v43_financial_reporting.sql`
Auto-posting Schema	`db/schema_v44_auto_financial_flows.sql`

Summary: All Financial Impacts at a Glance

#	Financial Impact	Trigger	Debit	Credit	Auto-post	Blocking
1	Sales Revenue	Sales log submitted	AR	Sales Rev + VAT	Yes	No
2	Payroll Accrual	Month-end batch	Salary Exp	Salary Pay	Yes	No
3	Advance Disbursement	Advance approved	Emp Advances	Bank/Cash	Yes	No
4a	Asset Capitalize	Asset assigned	Fixed Assets	Bank/AP	Yes	No
4b	Asset Depreciation	Monthly batch	Depr Exp	Accum Depr	Yes	No
5a	EOSB Accrual	Monthly batch	EOSB Exp	EOSB Provision	Yes	No
5b	EOSB Settlement	Employee terminated	EOSB Prov + Exp	Bank/Cash	Yes	No
6	Leave Encashment	On-demand / termination	Leave Exp	Salary Pay	Manual	No
7	GOSI Payment	Manual recording	GOSI Payable	Bank/Cash	Manual	No
8	Invoice Issuance	Invoice issued	AR	Sales Rev + VAT	On status change	No
9	Expense Recording	Expense approved	Exp Cat + VAT	AP / Bank	On approval	No
10	Payment Recording	Payment submitted	Varies by type	Bank/Cash	On submit	No